Wooah Choi

wooah.choi2@gmail.com | +44 07922 651195 | linkedin.com/in/wooah-choi | github.com/hncpyj | wooahchoi.com | backenddeveloper.tistory.com | London, UK | No visa sponsorship required until 2029

MSc AI at Queen Mary University of London, dissertation applying mechanistic interpretability to a PPO-trained causal Transformer in partially observable RL: validation-selected checkpoint reached Smin=0.959; causal interventions (retrieval blocking, opposite-memory patching) confirmed cue-dependent branch control. Former Software Engineer at Konan Technology, cutting Supreme Court litigation search latency from 5+ minutes to 8 seconds. Co-founder of GridFlow Trade (QMUL QIncubator), UK energy price forecasting pipeline achieving MAE £4.10/MWh (R^2=0.943), 76% error reduction. Experienced across the full ML stack: NLP/BERT, XGBoost, LSTM, Transformer architectures, and PyTorch-based RL.

EDUCATION

***MSc in Artificial Intelligence* | Queen Mary University of London,** London, UK (Sep 2025 - Present)

● Dissertation: Causal Cue Retrieval and Greedy Route Execution: A Mechanistic Analysis of a Transformer Policy in a Fixed MiniGrid T-Maze
● Core Modules: Machine Learning, Reinforcement Learning, Neural Networks and NLP, Information Retrieval, Conversational Agents and Dialogue Systems

Bachelor of Science in Computer Science (Double Major: Statistics and Data Science) | Korea National Open University, Seoul, South Korea (Mar 2021 - Aug 2023)

● Core Modules: Discrete Mathematics (A+), Linear Algebra (A+), Deep Learning (A)

Bachelor of Environmental Engineering & Digital Media Engineering | Catholic University of Korea, Seoul, South Korea (Mar 2013 – Feb 2018)

SKILLS

ML/AI: PyTorch, TensorFlow, Keras, scikit-learn, XGBoost, LightGBM, BERT, Transformer, Hugging Face, LLM, RAG, Embeddings, Generative AI, RL (PPO)

Data: pandas, NumPy, SQL, Oracle, MySQL, PostgreSQL

Backend & DevOps: Spring MVC, NestJS, RESTful API, GraphQL, Docker · Jenkins, GitLab CI/CD, AWS (EC2, S3, RDS), Linux

WORK EXPERIENCE

Jan 2022 – Present **Freelance ML/Software Consultant (Part-time)**

● Provided technical consulting for student research projects and software bug fixes on a project basis

Nov 2020 - Jun 2023 **Konan Technology Inc,** Seoul, South Korea

Software Engineer at AI Solution Development Team

Delivered production search and NLP systems across 4 enterprise clients (Supreme Court, Millie's Library, Heungkuk Insurance, Ministry of SMEs), owning full lifecycle from requirements analysis and infrastructure design through deployment and post-launch stabilization.

● **Supreme Court of Korea** [Java, Spring, Jenkins, GitLab, Python, Linux, Docker, C++, PyTorch, BERT, Oracle]

Built a next-generation electronic litigation search and analysis system, reducing search latency from 5+ minutes to 8 seconds and resolving 375+ misanalysed cases.

- Optimised Java backend algorithms and indexing pipeline to achieve 37x search speed improvement
- Configured Jenkins CI/CD pipelines on GitLab for unit/integration testing and production deployment
- Fine-tuned BERT-based NER model with SFX (dialogue behaviour analysis) and SRL (semantic role labelling) for query intent classification, achieving 4%+ accuracy improvement
- Extracted entities (case numbers, law names, party names) and applied semantic/vector search for query expansion and typo correction

- Extracted related/recommended keywords via LDA and time-series analysis of search logs
- Built search analytics dashboards (line, bubble, dendrogram charts) for administrators

● **Millie's Library** [C++, Python, MySQL]

- Engineered automated indexing system handling 100M+ data points, scalable to 1B+
- Improved search ranking quality by 38% via custom scoring logic
- Implemented auto-completion, related-term suggestion, and typo correction
- Delivered time-series analysis of search terms for recommended/popular keyword features

● **Heungkuk Fire & Marine Insurance** [Java, JavaScript, Oracle]

- Full-stack development of the web application's search system
- Implemented AJAX-based continuous scrolling for smooth user experience
- Built auto-completion and popular search term features

● **Ministry of SMEs and Startups** [Java, PostgreSQL]

- Developed search engine and RESTful API with text extraction and indexing from AWS S3
- Implemented recursive file/folder deletion logic for efficient data management

PROJECTS

Feb 2026 - Present **Causal Cue Retrieval and Greedy Route Execution: Mechanistic Analysis of a Transformer Policy in a POMDP T-Maze**

MSc Dissertation, QMUL · Supervisor: Paulo Rauber

● Studied whether a causal Transformer policy can retrieve a no-longer-visible instruction cue from within-episode context, compared against MLP and LSTM baselines trained with PPO in a partially observable MiniGrid environment.
● Designed a controlled ambiguous-memory task with symmetric layout and verified absence of visual leakage; introduced balanced cue success metric min(key, ball) to expose one-sided shortcut policies hidden by aggregate performance.
● Validation-selected checkpoint reached Smin=0.959; applied retrieval blocking and opposite-memory patching to confirm cue-dependent branch control; parameter-matched LSTM (h=304, Smin=0.898) also solved the task, shifting the contribution to mechanistic analysis rather than architecture comparison.
● Developed a reusable evaluation framework: branch-level failure decomposition, locked checkpoint selection, and causal context ablation taxonomy, separating genuine memory use from shortcut behaviour.

Nov 2025 – Apr 2026 **[GridFlow Trade] AI-Powered BESS Trading Platform** *(Co-Founder, AI & Backend)*

https://www.gridflowtrade.com

● Selected for QMUL QIncubator; pitched to investor panel, Apr 2026
● Engineered 48 domain-driven features (cyclical temporal encodings, price lags 30-min to 7-day, rolling statistics, weather × generation cross-features, LOLP, de-rated margin) on 55,000+ real UK half-hourly settlement periods via proprietary API
● Benchmarked Naive → Linear/Ridge → LightGBM → XGBoost → 2-layer LSTM with time-based train/test split; XGBoost achieved MAE £4.10/MWh (R^2=0.943) vs naive baseline £17.13/MWh — 76% error reduction
● Applied ADF stationarity tests, STL seasonal decomposition, ACF/PACF analysis, and leakage-free permutation importance for feature validation
● Built real-time data pipeline integrating NESO & Elexon APIs; deployed XGBoost quantile model to production generating automated charge/discharge/hold signals across 48 daily settlement periods

Apr 2025 **Stock AI YouTube Pipeline** *(Personal Project)*

● Automated weekly AI stock analysis videos using yfinance, OpenAI GPT-4, moviepy, and Stable Diffusion; pipeline fetches data, generates narrated scripts, synthesises visuals, and uploads to YouTube

Feb 2025 – Apr 2025 **BuildU** *(Next.js 15, OpenAI GPT-4, Tesseract.js, Vercel)*

● AI-powered university application advisor: analyses uploaded documents, extracts text via OCR, and generates personalised application strategies

- Integrated OCR via Tesseract.js for document parsing and OpenAI GPT-4 for contextual feedback generation
- Deployed on Vercel with optimised server-side rendering for fast load times

Aug 2024 **Invntz Hackathon** *(Spring Boot, Selenium, Jsoup)*

- Built a web scraping and aggregation platform collecting product data from 5 e-commerce sites via automated browser crawling
- Awarded 2nd place at the Invntz hackathon

Jul 2024 - Oct 2024 **Generative AI-Based Storytelling and Illustration Agent System**

- Built an end-to-end pipeline that converts keyword inputs into narrative stories using generative AI APIs
- Used generated stories as prompts for image-generation APIs to create illustrations for digital picture books

Jul 2023 – Jan 2024 **ATE — Language Learning App** *(React Native/Expo, NestJS, Prisma, PostgreSQL)*

- Developed a language learning mobile app with spaced repetition and AI-generated practice sentences
- Implemented backend REST API with NestJS, Prisma ORM, and PostgreSQL, deployed on AWS EC2

Jul 2022 - Oct 2022 **[Kaggle] RSNA 2022 Cervical Spine Fracture Detection**

- Applied data augmentation, resizing, normalization, and segmentation techniques to preprocess images, identifying cancerous regions through contouring and drawing bounding boxes to extract ROI
- Predicted seven cervical spine fracture probabilities and vertebra presence per slice using EfficientNetV2

May 2022 - Aug 2022 **[Kaggle] Google AI4Code – Understand Code in Python Notebooks**

- Improved cell execution order predictions by applying transformer-based models (CodeBERT, DistilBERT) with an optimized ensemble approach (0.748:0.252)
- Conducted EDA, feature engineering, and iterative experiments, leveraging research insights to refine accuracy and validate results

ACADEMIC SERVICE

Aug 2026 **Reviewer, FinNLP 2026 @ EMNLP 2026** *(11th Workshop on Financial Technology and Natural Language Processing, ACL SIG-FinTech)*

REFERENCES

Available upon request